



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title :

Kagojer Stup: Geo-Prioritized Multilingual News Retrieval and Content Management System

Group No : 11, CSE370 Lab Section : 02, Spring 2026		
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Introduction

Kagojer Stup is a geo-prioritized multilingual news retrieval and content management system designed to deliver highly relevant, trustworthy, and personalized news experiences. The platform leverages real-time geographic context to rank news, ensuring users first see **local updates**, followed by national and international coverage.

Example: A user in Dhaka will see nearby issues like waterlogging in Dhanmondi or metro rail updates before global economic news.

The system features a **geographically prioritized keyword search**, organizing results into Local → National → International categories while maintaining keyword relevance.

Example: Searching “election” will show Dhaka-specific election updates first, then Bangladesh-wide coverage, and finally international elections.

A key innovation is the **hybrid credibility scoring system**, which combines multi-source verification with **user participation**. News reported by multiple trusted outlets receives higher scores, while users can vote articles as “*Real*” or “*Fake*”.

Example: If a story is covered by Prothom Alo and The Daily Star and receives mostly “Real” votes, it may display a 90%+ trust score, whereas conflicting or flagged news ranks lower.

The platform also includes a **live news stream**, where articles are continuously updated with region tags, timestamps, and trust scores.

Example: Political news in Dhaka appears with “100% Trust Score” if verified by multiple sources.

To improve accessibility, **AI-powered summaries and translations** allow quick understanding.

Example: A long English article can be summarized and translated into Bangla instantly.

The platform also features an advanced **personalized news feed** that continuously adapts to individual user behavior and preferences. The system analyzes multiple interaction signals such as **clicked articles, reading time, search history, followed sources, and engagement actions (e.g., shares, bookmarks)** to understand user interests over time. Based on this, it dynamically reorders and recommends content. Also in this section, the platform allows users to **follow specific news portals**, ensuring their preferred sources appear more frequently in the personalized feed.

Example: If a user frequently reads political news from Dhaka and follows sources like Prothom Alo, their feed will prioritize local political updates from those outlets. Similarly, if another user engages more with business or technology news, those categories will appear more prominently in their feed.

User engagement is enhanced through a **real-time trending section**.

Example: “Top 10 Trending” highlights widely read or shared stories within 24 hours.

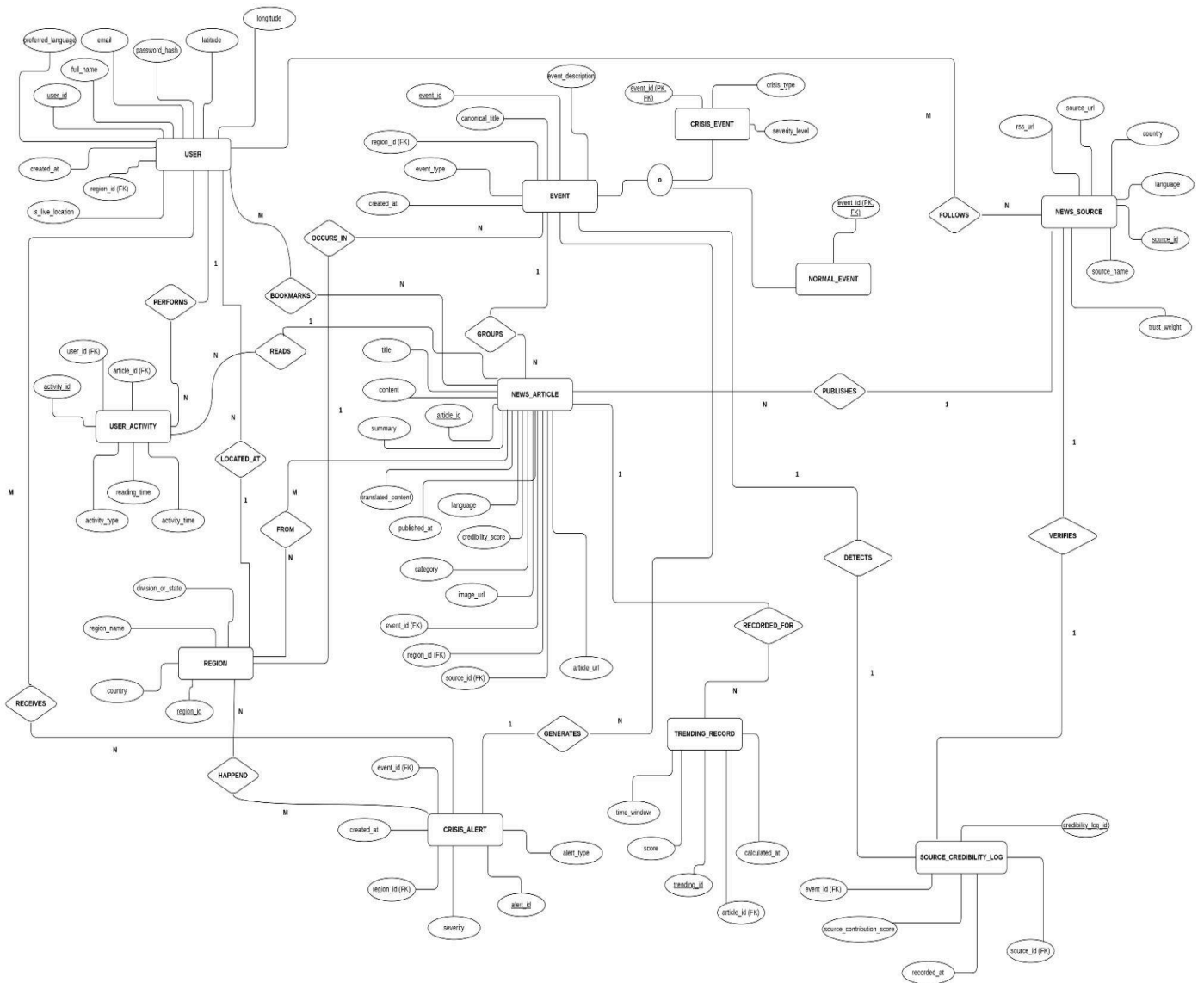
Finally, **crisis alerts** notify users of nearby emergencies, and an **interactive news map** allows exploration via location pins.

Example: Clicking a red pin in Sylhet reveals recent accident or weather-related news.

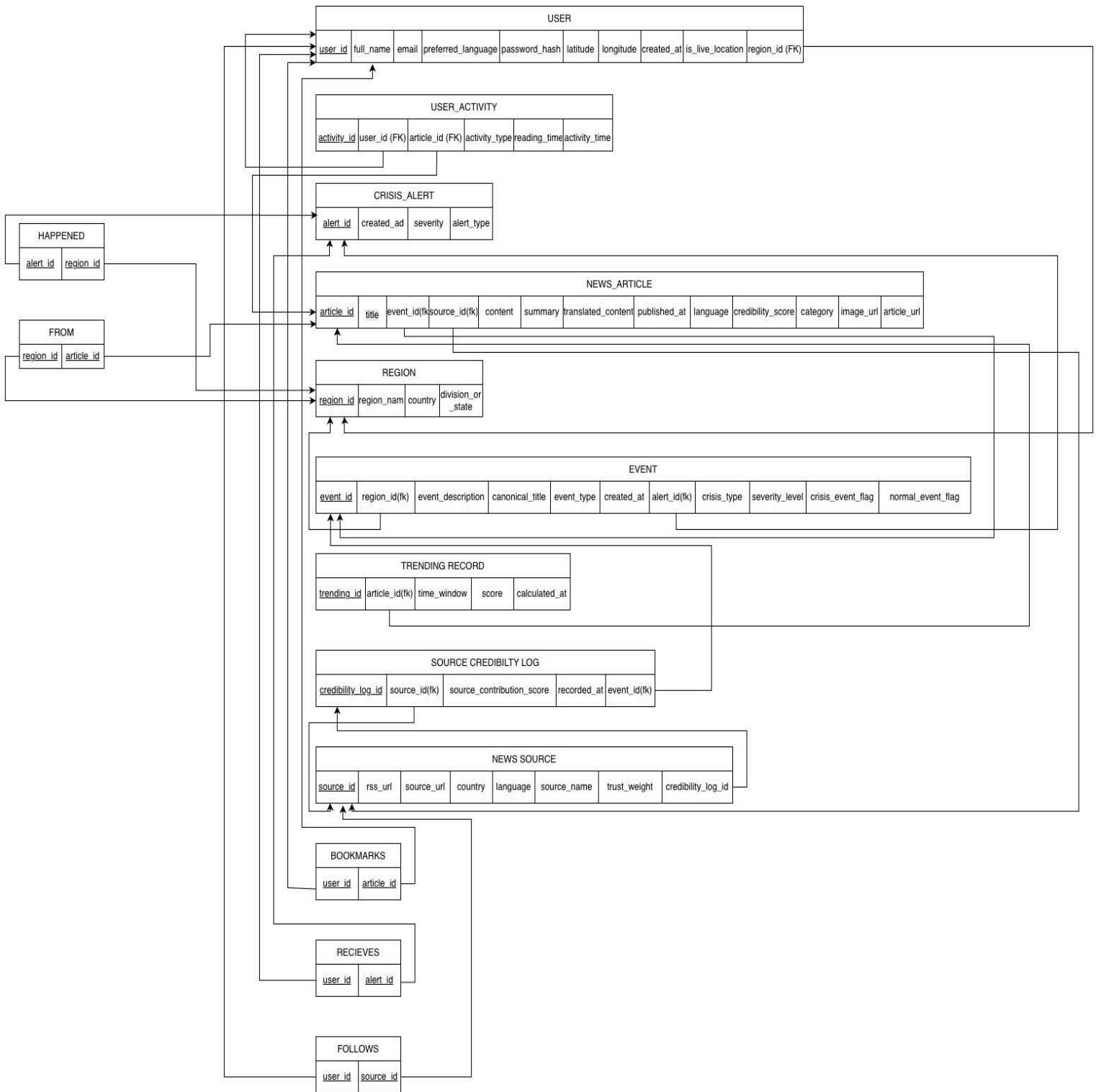
Project Features

ID, Name	Features [3 per member]	
24101524, MD. Abdullah Al Jarif	Ft 1	Region-Based News Ranking
	Ft 2	Geographically Prioritized Result Based on Searched Keyword
	Ft 3	Source Credibility Scoring
24101136, MD. Mahfuz Ahmed Emon	Ft 1	Personalized News Feed
	Ft 2	AI News Summary & Translation + Categorization
	Ft 3	Crisis Alert Based on Live Location + Map
24101060, Mehrab Hossain Nizami	Ft 1	Real-Time Trending Detection
	Ft 2	Follow News Portals
	Ft 3	Location-Based Breaking News Map

ER/EER Diagram

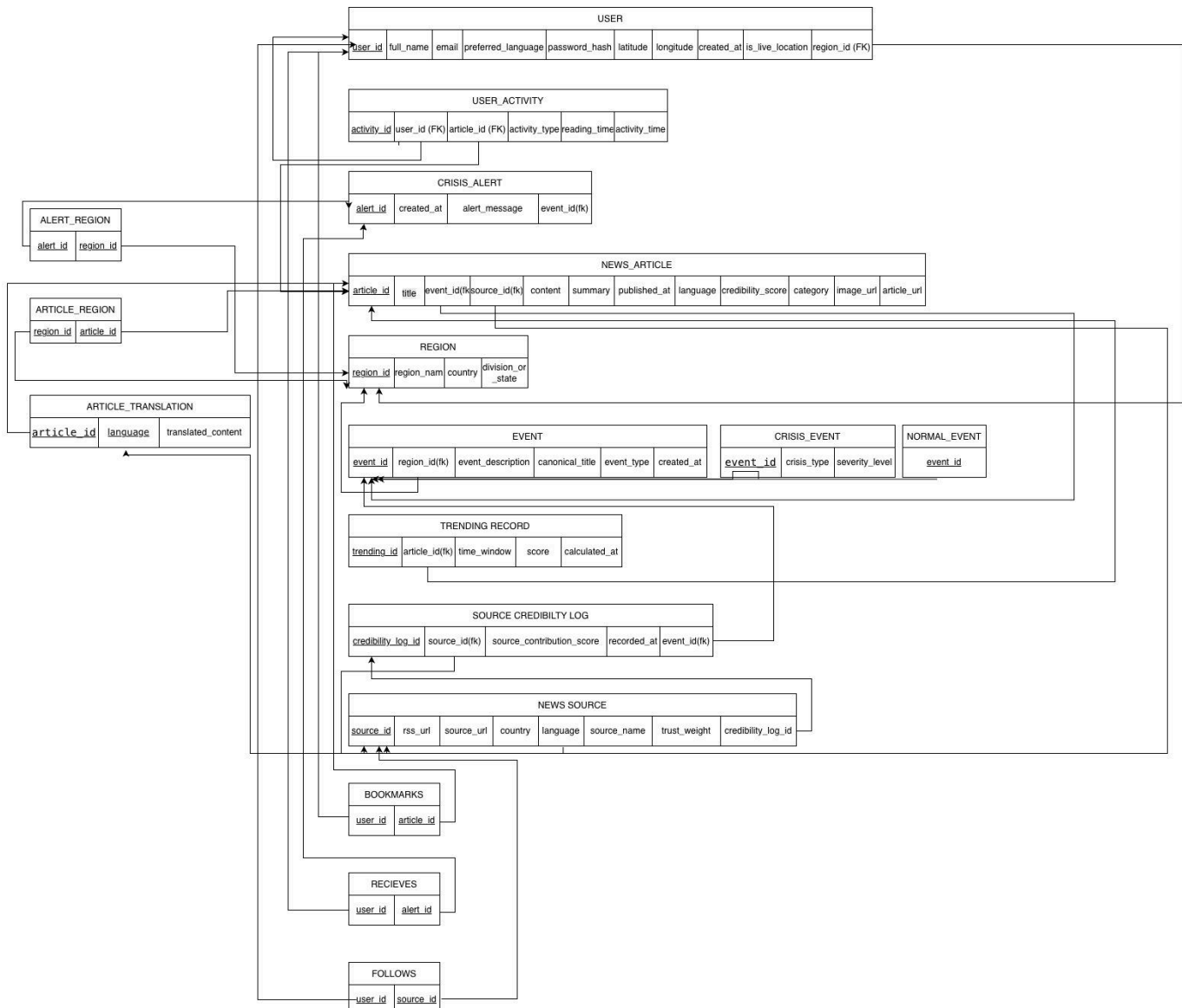


Schema Diagram



Normalization

- Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?



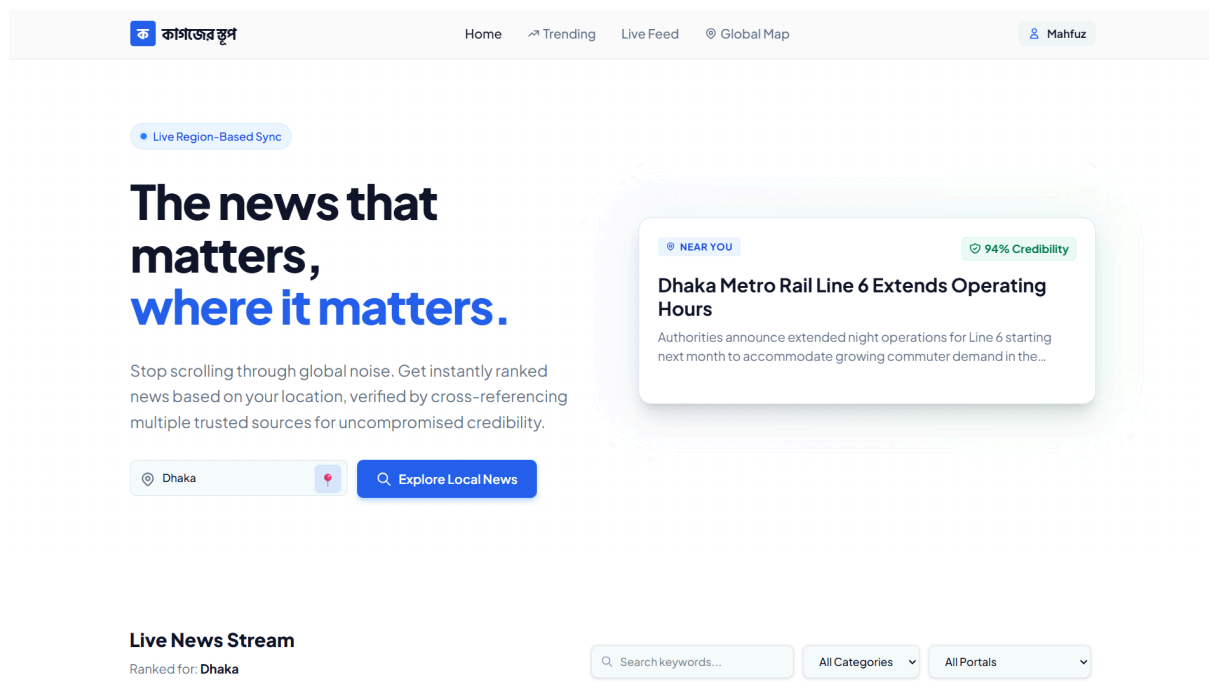
- Yes, the schema is in 1NF as there is no nested relationship, no multivalued or composite attributes.
- The converted schema is in 2NF as no partial dependency exists here and the schema is also in 1NF.
- The converted schema is in 3NF as no transitive dependencies exist here and also follows 2NF.

Frontend Development

Overview: The frontend of Kagojer Stup is built as a Single Page Application (SPA) using **React.js** and styled with **Tailwind CSS** for a responsive, modern "glassmorphism" aesthetic. Routing is handled via **react-router-dom**, and interactive mapping is implemented using **react-leaflet**. State management is heavily utilized to handle user sessions, dynamic filtering, and real-time UI updates without page reloads.

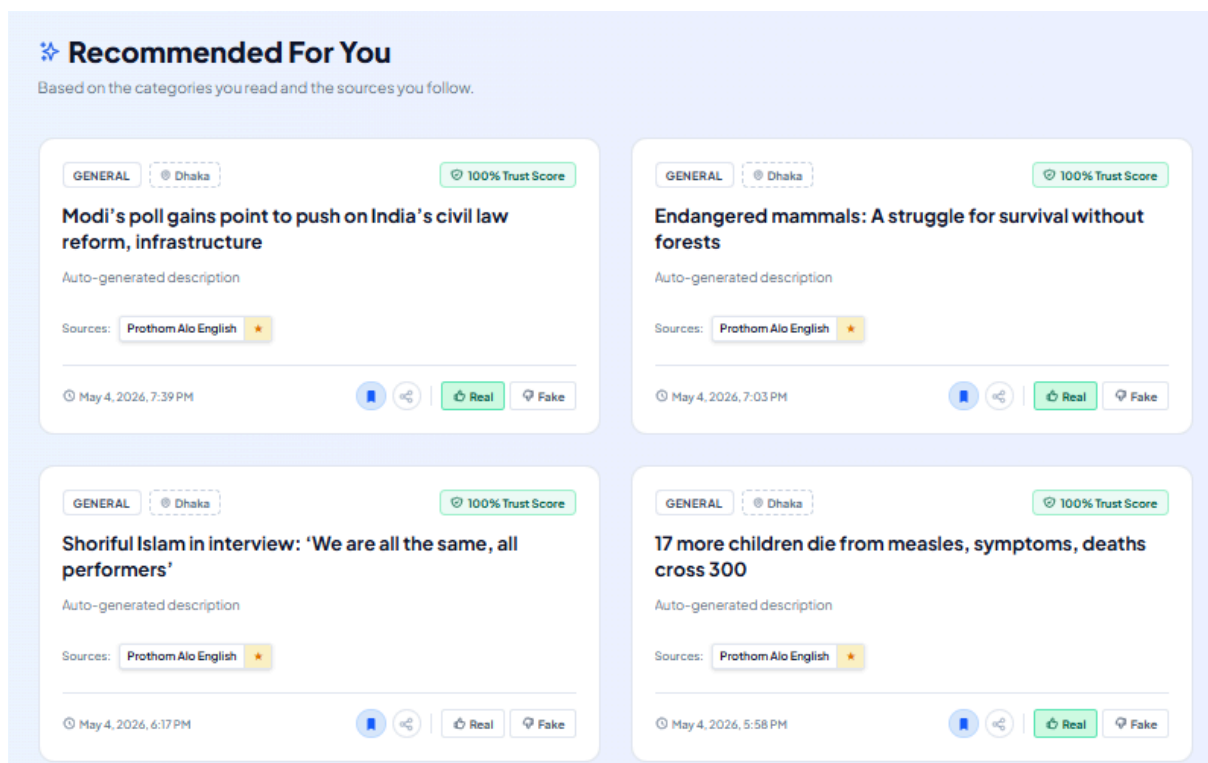
Contribution of ID: 24101524, Name: MD. Abdullah Al Jarif

- **Region-Based News Ranking & Search UI:** Implemented the primary Hero section interface, including the region-selection dropdown and keyword search bar. Engineered the **filteredFeed** logic in React that dynamically filters the **newsFeed** state array in real-time based on the selected region, category, news portal, and user search queries.
- **Dynamic Feed Rendering:** Managed the integration of the main "Live News Stream" section, ensuring it seamlessly displays standard cards or highlights dynamically shared single-article links using URL parameters (**useParams**).



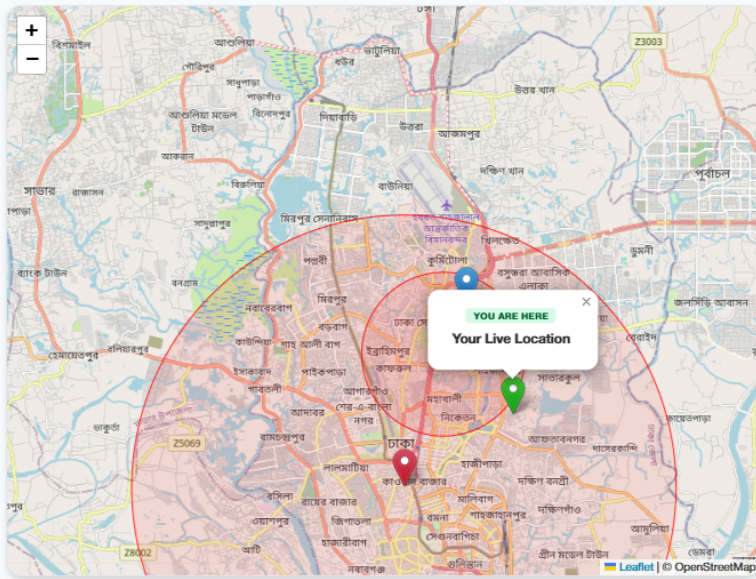
Contribution of ID: 24101136, Name: Md. Mahfuz Ahmed Emon

- **Personalized Feed & News Card Component:** Developed the `renderNewsCard` reusable component, which conditionally renders compact versions for the horizontal scrolling "Recommended For You" section and full versions for the live feed. Integrated UI elements for Source Credibility Scoring (displaying Trust Score badges).
- **AI Summary, Translation & Activity Tracking:** Built the interactive toggles for "Read AI Summary" and "Translate" within the news cards. Also implemented the frontend event handlers (`handleVote`, `handleBookmark`, `handleArticleClick`) that instantly update the UI optimistically while sending POST requests to the backend.
- **Crisis Alert Banner:** Created the `localCrisisAlert` sticky banner that pulses red and alerts users when the geolocation API detects they are inside an active crisis radius.
- *[Insert Screenshot: Personalized Horizontal Feed and a News Card showing Trust Score/AI Buttons]*



Interactive News Map

Click on a **blue pin** for breaking news, a **red pin** for active crises, and see the **green pin** for your live location.



No Region Selected

Click on any blue pin on the map to explore breaking news from that specific area.

POLITICS

Dhaka

100% Trust Score

Sara Hossain Criticises Bail Denials in July Uprising Cases, Raises Human Rights Concerns

Senior Supreme Court lawyer Sara Hossain criticized the repeated denial of bail in cases related to the July mass uprising killings, highlighting the difficulty judges face in granting bail even without presented evidence. She emphasized that human rights are universal and should apply equally to everyone, irrespective of political ideology, expressing concern that bail is being denied for political purposes, which she views as a mockery of justice.

Hide Summary

Translate

Sources: Prothom Alo English

May 4, 2026, 11:30 AM

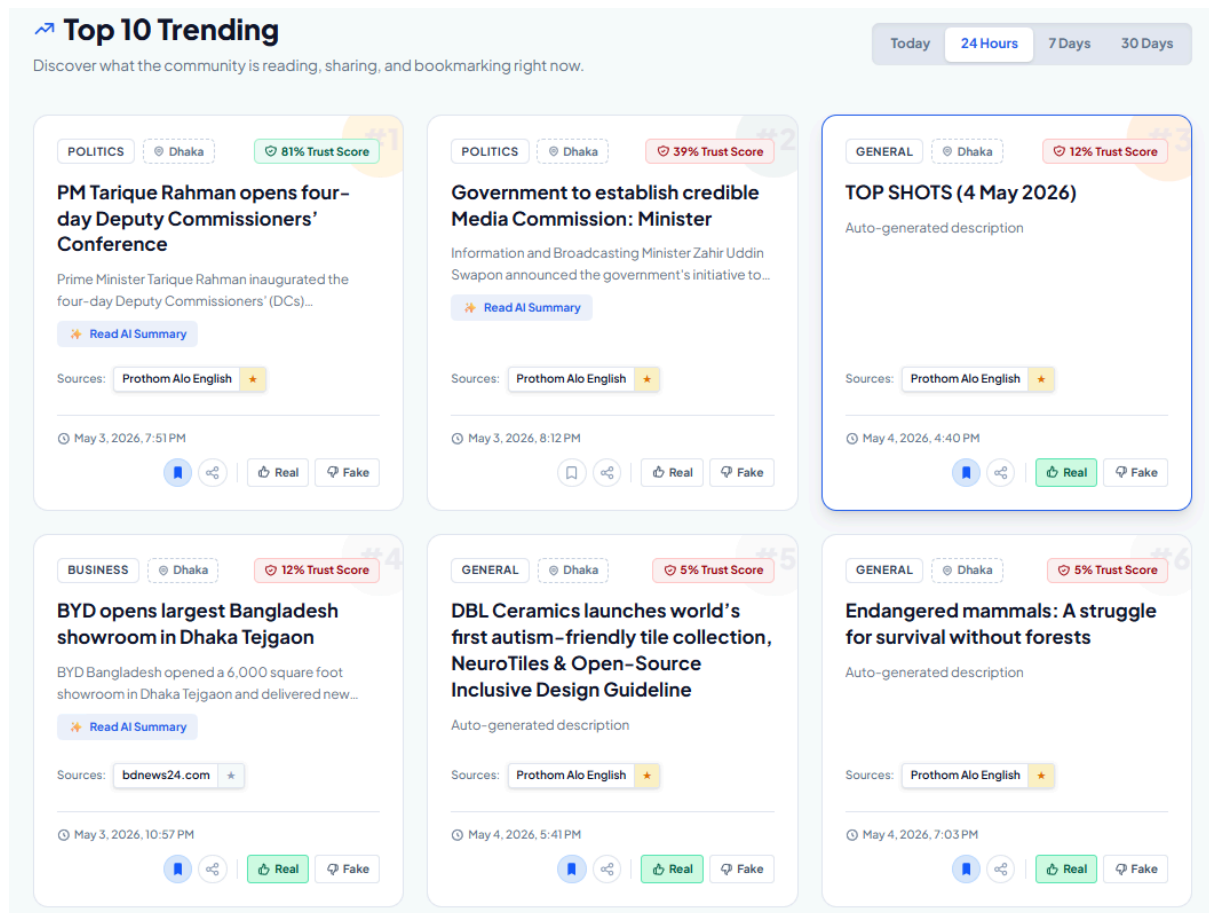


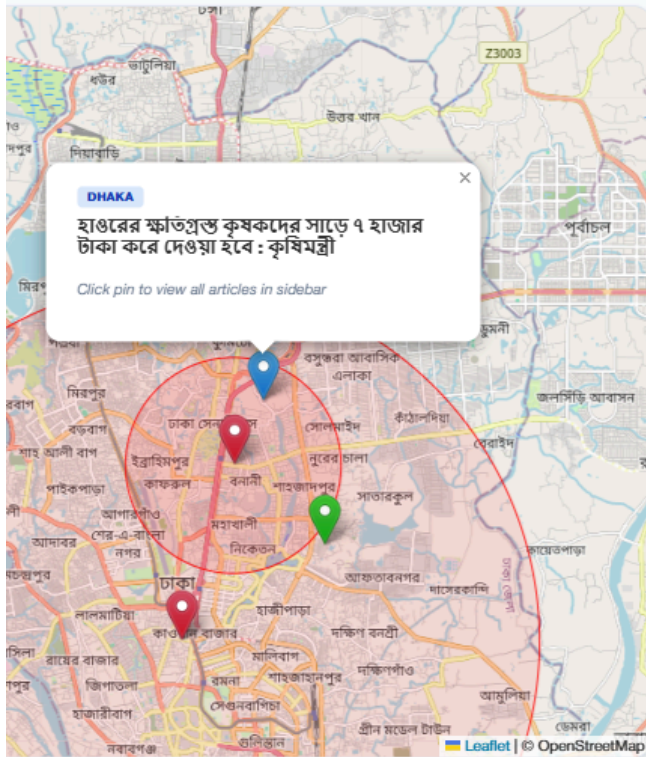
Real

Fake

Contribution of ID: 24101060, Name: Mehrab Hossain Nizami

- **Location-Based Breaking News Map:** Implemented the **MapContainer** using Leaflet, plotting custom SVG icons (Blue for News, Red for Crises, Green for User Location) based on geographic coordinates. Built the interactive map sidebar that displays a customized news feed when a specific region pin is clicked.
- **Real-Time Trending Dashboard:** Developed the "Top 10 Trending" UI section, including the time-window toggle buttons (Today, 24 Hours, 7 Days, 30 Days) that dynamically fetch and display ranked news cards with custom ranking badges.
- **User Dashboard & Follows:** Designed the **UserProfile** component, displaying saved bookmarks, recent activity logs, and followed news portals. Integrated the "Follow" (★) button logic across the application.
- *[Insert Screenshot: Interactive Leaflet Map and Trending News Section]*





VIEWING NEWS FOR

Dhaka

77 updates

GENERAL

May 4, 2026, 8:36 PM

হাওরের ক্ষতিগ্রস্ত কৃষকদের সাড়ে ৭ হাজার টাকা করে দেওয়া হবে: কৃষিমন্ত্রী

Auto-generated description

Source: Kaler Kantho

[Read Full](#)

GENERAL

May 4, 2026, 8:30 PM

নাপোলির কাছ থেকে শিরোপা পুনরুদ্ধার ইন্টারের

Auto-generated description

Source: Kaler Kantho

[Read Full](#)

GENERAL

May 4, 2026, 8:29 PM

ইবিতে বসুন্ধরা শুভসংঘের আয়োজনে সাহিত্য আড্ডা অনুষ্ঠিত

Auto-generated description

Source: Kaler Kantho

[Read Full](#)

Backend Development

Overview: The backend of Kagojer Stup is a robust RESTful API built with Node.js and Express.js. It connects to a MySQL relational database using the mysql2/promise connection pool for efficient, asynchronous queries. The architecture is designed to handle complex SQL aggregations, geographic distance calculations, secure user authentication, and seamless AI integration.

Contribution of ID: 24101524, Name: MD. Abdullah Al Jarif

- **Core News Retrieval Endpoints:** Developed the primary `/api/news` GET endpoint. This endpoint executes complex SQL queries combining JOIN statements across the `news_article`, `event`, `region`, and `news_source` tables. It handles the foundational data delivery to the frontend, incorporating dynamic WHERE clauses for region-based filtering and ORDER BY for chronological sorting.
- **Authentication & User Management:** Built the secure `/api/signup` and `/api/login` POST endpoints. Implemented password hashing using the bcrypt library to ensure user data security before insertion into the `app_user` table. Handled JWT/Session creation logic to manage user states.
- **Source Credibility Scoring:** Integrated sub-queries into the news retrieval endpoints to calculate the article `base_score`. This was achieved by averaging source trust weights and adjusting for multi-source verification (counting distinct `source_ids` linked to a single `event_id`).

```
// API Endpoint: Get localized news feed (Public)
app.get('/api/news', async (req, res) => {
  try {
    const { region } = req.query;

    let sqlQuery = `
      SELECT
        na.article_id AS id,
        na.title,
        na.article_url AS url,
        na.summary,
        na.translation,
        na.category,
        na.published_at AS time,
        r.region_name AS region,

        (
          SELECT LEAST(100, ROUND(AVG(ns2.trust_weight) * 10 + (COUNT(DISTINCT na2.source_id) - 1) * 5))
          FROM news_article na2
          JOIN news_source ns2 ON na2.source_id = ns2.source_id
          WHERE na2.event_id = na.event_id
        ) AS base_score,

        (SELECT COUNT(*) FROM user_activity ua WHERE ua.article_id = na.article_id AND ua.activity_type = 'vote_real') AS real_votes,
        (SELECT COUNT(*) FROM user_activity ua WHERE ua.article_id = na.article_id AND ua.activity_type = 'vote_fake') AS fake_votes,

        (
          SELECT GROUP_CONCAT(DISTINCT ns3.source_name SEPARATOR '||')
          FROM news_article na3
          JOIN news_source ns3 ON na3.source_id = ns3.source_id
          WHERE na3.event_id = na.event_id
        ) AS all_event_sources
    `
```

Contribution of ID: 24101136, Name: Md. Mahfuz Ahmed Emon

- **AI Worker Integration (Summarization, Translation & Categorization):** Developed the `/api/trigger-ai` POST endpoint and managed the `aiWorker.js` integration. This backend module triggers automated processing of newly fetched articles, utilizing the Gemini AI API to generate concise summaries, translate English content into Bangla, and dynamically classify articles into specific categories before updating the database.
- **Personalization & Recommendation Engine:** Engineered the advanced `/api/news/personalized` GET endpoint. This endpoint dynamically calculates a `personalSortWeight` for each article. The logic queries the `user_activity` table to determine category affinity, adds flat point boosts for sources found in the `follows` table, and applies a mathematical time-degradation penalty for older articles. Implemented a strict mathematical cap (`Math.min`) to prevent algorithmic monopolization.
- **Proximity Crisis Detection:** Developed the `/api/check-crisis-alert` POST endpoint utilizing the Haversine formula within the SQL query to calculate the exact distance between the user's live GPS coordinates and active crisis events in the database to trigger targeted alerts.

```
// API Endpoint: Get Personalized News Feed (Logged In - AI Recommended)
app.get('/api/news/personalized', async (req, res) => {
  try {
    const { userId } = req.query;

    if (!userId) {
      return res.status(400).json({ error: 'User ID required' });
    }

    // 1. Calculate Category Affinity (What topics do they like?)
    const [categoryPrefs] = await pool.execute(`
      SELECT na.category, COUNT(*) as weight
      FROM user_activity ua
      JOIN news_article na ON ua.article_id = na.article_id
      WHERE ua.user_id = ?
      GROUP BY na.category
      ORDER BY weight DESC
    `, [userId]);

    const preferredCategories = categoryPrefs.reduce((acc, curr) => {
      acc[curr.category] = curr.weight;
      return acc;
    }, {});
```



```
// 2. Function to ask Gemini to format it
async function analyzeWithGemini(articleText) {
  const model = genAI.getGenerativeModel({ model: "gemini-2.5-flash" });

  const prompt = `
  You are an expert news analyst and translator.
  Read the following article and extract the data strictly as a raw JSON object.
  Do NOT wrap the response in markdown blocks (like \`\`\`json). Just output the raw JSON data.

  Rules for translation:
  - If the article is written in English, translate your summary into Bengali.
  - If the article is written in Bengali, translate your summary into English.

  Format:
  {
    "title": "Main headline",
    "summary": "A strict 2-sentence summary in the original language of the article",
    "translation": "The 2-sentence summary translated into the other language (English to Bengali OR Bengali to English)",
    "category": "Choose one: Politics, Weather, Tech, Crime, Business, Sports, Local, General",
    "region_name": "The specific district or city (e.g., 'Dhaka', 'Sylhet', 'Chattogram'). If nationwide or international, output 'All'"
  }

  Article Text:
  ${articleText}
  `;

  const result = await model.generateContent(prompt);
  let rawText = result.response.text();

  rawText = rawText.replace(/```json/gi, '').replace(/```/gi, '').trim();
}
```

Contribution of ID: 24101060, Name: Mehrab Hossain Nizami

- **RSS News Fetcher Implementation:** Designed and implemented the automated news fetcher module (`newsFetcher.js`), which connects to multiple RSS feeds to continuously scrape and ingest the latest articles and breaking news directly into the database.
- **Trending News Engine:** Developed the `/api/trending-news` GET endpoint. Wrote the SQL grouping logic that calculates a `trending_score` by assigning different weights to specific user actions (e.g., Clicks = 1, Reads = 2, Bookmarks = 4, Shares = 5) stored in the `user_activity` table. Filtered results based on dynamic INTERVAL time windows (24h, 7d, 30d).
- **Map Data Aggregation & Activity Tracking:** Created the `/api/map-news` GET endpoint using JavaScript reduce functions on the backend to group flat SQL rows into structured, nested JSON objects categorized by `region_name` for the frontend Leaflet map. Additionally, built the `/api/track-activity`, `/api/vote`, `/api/bookmarks`, and `/api/follows` endpoints, handling user interactions and implementing logic to delete "ghost data" to maintain the accuracy of the personalization algorithm.

```

import cron from 'node-cron';
import Parser from 'rss-parser';

// THE FIX 1: Set up the parser with a "Chrome Browser" disguise to bypass firewalls
const parser = new Parser({
  timeout: 15000, // Give up after 15 seconds so one bad site doesn't freeze the loop
  headers: {
    'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/124.0.0.0 Safari/537.36',
    'Accept': 'application/rss+xml, application/xml, text/xml; q=0.1'
  }
});

export const startNewsFetcher = (pool) => {

  cron.schedule('*/*/* * * * *', async () => {
    console.log(`🕒 [CROW] Starting 30-minute news fetch cycle...`);

    try {
      // 1. Get all active RSS feeds
      const [sources] = await pool.execute(
        "SELECT source_id, source_name, rss_url FROM news_source WHERE rss_url IS NOT NULL AND rss_url != ''"
      );

      for (const source of sources) {
        console.log(`📡 Requesting feed for: ${source.source_name}...`);

        try {
          // THE FIX 2: Use AllOrigins as a trusted middleman to bypass IP blocks
          const proxyUrl = `https://api.allorigins.win/raw?url=${encodeURIComponent(source.rss_url)}`;
          const feed = await parser.parseURL(proxyUrl);

          let newArticlesCount = 0;

          for (const item of feed.items) {

```


Source Code Repository

<https://github.com/imooooooooon/KagojerStup>

<https://github.com/imooooooooon/CSE370-Project>

<https://kagojer-stup.vercel.app/>

Conclusion

In the end, **Kagojer Stup** wasn't just a project, it was a full-blown adventure. What started with neatly planned EER diagrams, schemas, and normalization (beautifully crafted in draw.io, of course) quickly turned into a survival game against bugs, breakdowns, and pure confusion. The front end had its own personality. Sometimes the entire page would just turn white, as if it had given up on life. Other times, the layout would collapse, elements would escape their containers like rebellious teenagers, and refreshing the page would magically erase all previous activity (a feature we definitely didn't intend).

The backend wasn't any calmer. User data would occasionally disappear into the void, or worse, come back with completely wrong information, like it had its own imagination. And just when things started to make sense, the real villain appeared: running out of prompt limits on the AI assistant right when it was needed the most.

But through all the chaos, debugging sessions, and "why is this happening?" moments, it turned out to be a genuinely fun and rewarding journey. We learned more than expected, not just about systems and coding, but about patience and persistence.

Now, Kagojer Stup stands as more than just a project. It's a foundation. We aim to take it much further, refining it into something impactful, something people across the nation can actually use and rely on. Because after everything, if it doesn't help real users, then all those white screens were for nothing.

References

ChatGpt, Gemini, Railway, Vercel, Render, Bux